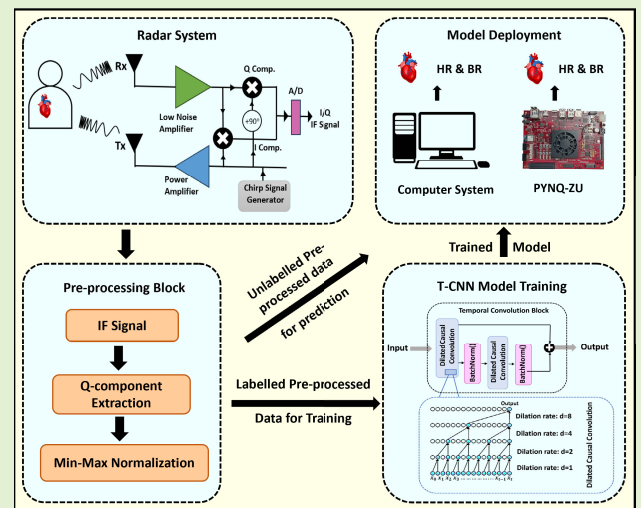


Contactless Vital Sign Monitoring on Edge Devices With Low-Complexity T-CNN and mmWave FMCW Radar

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Abstract—Noncontacted remote monitoring of vital signs has become a key area of interest in health research, providing a safer approach during epidemic outbreaks when limited contact is essential to prevent infection. Hence, this study introduces an efficient method for monitoring vital signs utilizing millimeter frequency-modulated continuous-wave (FMCW) radar data. Two temporal convolutional neural networks (T-CNNs) are designed and implemented to accurately estimate heart rate (HR) and breath rate (BR) from FMCW radar time-series data. These models achieve high precision with minimal preprocessing, eliminating the need for traditional range bin identification and respiratory harmonic filtering, streamlining the vital sign estimation process while maintaining low average mean absolute error (MAE). By employing temporal convolutions (TCs), the proposed models expand their receptive field without substantially increasing parameter count, optimizing it for real-time applications. The proposed models are compared with the state-of-the-art models, viz. ResNet1D and T-LeNet, and they were found to achieve a lower MAE. The designed models were successfully implemented on a high-performance computing system and the system-on-chip (SoC) platform PYNQ-ZU, achieving notable performance with an average MAE of 2.37 beats per minute for HR and 1.18 breaths per minute for BR. On PYNQ-ZU, the models achieved an impressive inference time of 12.6 and 6.1 ms for HR and BR, respectively. With computational efficiency on a portable device and without lowering accuracy, this approach enables real-time applications, unlocking diverse practical opportunities in healthcare.

Index Terms—77 GHz, breath rate (BR), frequency-modulated continuous-wave (FMCW), healthcare, heart rate (HR), mmWave radar, PYNQ-ZU, temporal convolutions (TCs).



I. INTRODUCTION

MONITORING vital signs like heart rate (HR) and respiration rate or breath rate (BR) have become particularly essential, as these metrics are crucial indicators

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of health status [1], [2], [3]. Traditionally, monitoring these signs has relied on methods such as photoplethysmography (PPG) [4] and electrocardiography (ECG) [5] that are contact-based. However, these approaches can pose discomfort or additional risk for certain patients, such as those with burn injuries. To mitigate these challenges, noncontact radar-based technologies have gained attention for its accuracy and noninvasive nature [6], [7], [8], [9], [10], [11], [12], [13], [14]. Frequency-modulated continuous-wave (FMCW) radars, in particular, are well-suited for detecting vital signs [15], [16], [17], [18] when compared to continuous wave as it captures cleaner and more accurate signals, which helps in analyzing heart activity more precisely, including subtle details like heart sounds and variability [19]. Unlike CW radar, it works well at different distances and frequencies without needing constant recalibration. Additionally, it enables multitarget tracking with spatial localization [20], [21], [22], [23], [24], making it ideal for applications like rescue

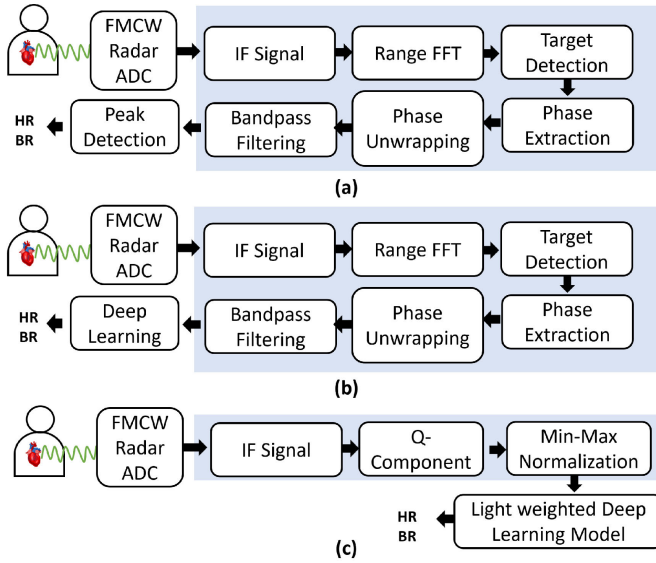


Fig. 1. Comparison of approaches for vital sign estimation: (a) Traditional signal-processing workflow. (b) General framework integrating deep learning. (c) Proposed low-complexity deep learning approach for HR and BR estimation.

missions and tele-health [19]. Similarly, ultrasound-based vital sign monitoring, another contactless technique, is highly sensitive to orientation; even a slight head/body tilt greatly affects the signal [25], whereas FMCW radar has shown better performance even under motion scenarios and varying orientations [18], [26].

Furthermore, mmWave FMCW radar's compact size [27] and low power consumption make it ideal for unobtrusive, contactless health monitoring systems, particularly in home-based care and continuous remote patient monitoring environments. These capabilities have already proved effective in applications like infant health tracking [28], [29], sleep monitoring [30], [31], posture classification [32], contactless ECG monitoring through ECG reconstruction using deep learning [33], and gesture detection [34].

HR and BR are crucial metrics that offer valuable insights into health conditions. HR is typically expressed in beats per minute (bpm), whereas BR is expressed in breaths per minute (br/min). Radar systems can measure these vital signs by detecting small distance changes caused by heart contractions and relaxations in the chest and chest wall movement due to respiration. Using FMCW radar systems, the HR and BR are measured through the fluctuations in the phase of the selected range bin using a conventional signal-processing chain. The conventional signal-processing chain includes acquiring the intermediate frequency (IF) signal, extraction of the range bin by performing range fast Fourier transform (FFT), followed by extraction of phase, phase unwrapping, and finally, bandpass filtering, as shown in Fig. 1(a) [35]. The phase-unwrapped signal contains information for both HR and BR. Bandpass filtering is applied to extract these signals, with frequency ranges of 1–2 Hz and 0.1–0.5 Hz for HR and BR, respectively. After extraction, peak detection is performed to identify the times at which peaks occur. In peak detection method, suppose $t_{pd1}, t_{pd2}, \dots, t_{pdn}$ represent these peak times. The time intervals between consecutive peaks are computed as $\Delta t_{pdi} = t_{pdi+1} - t_{pdi}$ for $i = 1, 2, \dots, n-1$. The average period,

T_{pd} , is then calculated as the mean of these intervals: $T_{pd} = 1/n - 1 \sum_{i=1}^{n-1} \Delta t_{pdi}$. The frequency, f_{pd} , is the reciprocal of the average period: $f_{pd} = 1/T_{pd}$. The BR in br/min is given by $BR_{pd} = 60 \times f_{pd}$. The HR is computed in the same way, by detecting the peaks in the filtered phase unwrapped signal, expressed in bpm. Although traditional signal-processing chains have laid the groundwork for measuring vital sign signals, they often lack both accuracy and efficiency [36], [37]. Accurately identifying the correct range bin and filtering out respiratory harmonics completely is a complex and computationally expensive process [35], which can introduce errors and lead to unreliable results. As a result, many researchers have employed extensive preprocessing techniques to address these issues. However, these methods are often complex and difficult to scale for practical deployment. For instance, Liu et al. [38] used the digital filter to extract the HR and BR signals by suppressing the harmonics. Similarly, in another study, Chen et al. [37] addressed baseline drift from human micromotion using median filtering. They proposed a DR-MUSIC algorithm, combining RLS filtering with the MUSIC method, to suppress respiratory harmonics and separate heart and breathing signals. HR was then estimated through spectral peak search.

With the rapid advancements in AI, several studies [35], [36], [39], [40] have explored the estimation of vital signs using deep learning techniques, frequently incorporating extensive preprocessing steps. For instance, Chang et al. [40] proposed a weighted deep learning scheme for HR and BR estimation, but this also requires extensive preprocessing. The scheme proposed in [40] extracts the features twice, first using signal magnitude, frequency, and phase from each of the selected range bins and then these features are fed to a convolution block (network) that calculates the weights of these features and then inputs them to another CNN with around 1.05M (million) parameters, for HR and BR estimation. This approach was indeed effective, but increased the computation in feature extraction and these two neural networks, making it inefficient for real-time applications.

Furthermore, Kim et al. [36] used a complete conventional signal-processing chain, that is, extracting the range bin, unwrapping the phase signal, and filtering it before feeding the data into their CNN model as shown in Fig. 1(b). This essentially makes their CNN functions like a bandpass filter, without actually reducing the overall preprocessing workload. Similarly, another recent study [39] employed a two-stage zoom FFT (zFFT) to determine target distance and enhanced range resolution. The first FFT stage detects the target's distance and shifts the signal to baseband, while the second stage refines the distance detection for improved resolution. From the enhanced range profiles, multiple amplitude, phase, and frequency-based respiration and HR estimates are extracted. These multiple estimates are input into a CNN-based encoder-decoder network (zDAWS), which calculates the weights of HR and BR and fuses them with the phase sequence estimations obtained from the range indices to finally predict the HR and BR. While both studies [36], [39] achieved lower estimation errors, the heavy reliance on preprocessing substantially increased computation, posing a real barrier to efficient, real-time deployment on embedded hardware. This demands the requirement for approaches that balance

preprocessing needs with an efficient model for deployment on edge devices.

To address these limitations, this study introduces light-weight temporal convolutional neural networks (T-CNNs) for HR and BR estimation, designed to remove preprocessing requirements such as range bin identification and respiratory harmonic filtering, as shown in Fig. 1(c). These improvements make it easier to handle data while reducing computational requirements and, therefore, inference times. The models' design makes it extremely suitable for deployment on hardware platforms with resource constraints, such as the edge device, PYNQ-ZU, a system-on-chip (SoC). In addition, the T-CNN results in better performance metrics, such as lower mean absolute error (MAE) and root-mean-square error (RMSE), compared to traditional deep learning models running on high-performance computing systems. The proposed T-CNN models were evaluated by comparing with advanced time-series models such as ResNet1D [41], [42] and time-LeNet (T-LeNet) [43]. ResNet1D was chosen for its robust feature extraction capabilities, which stem from its strong residual block architecture. This architecture is well-suited for handling complex sequential data, making it a reliable standard for comparison, while the T-LeNet was included due to its lightweight design and demonstrated effectiveness in real-time applications, particularly in resource-constrained computational environments. Its efficiency and adaptability make it a suitable baseline for evaluating time-series analysis frameworks. These models form effective benchmarks for testing the exactness and efficiency of these time-series analysis frameworks.

The paper is structured as follows. Section II introduces the fundamentals of FMCW radar systems and their application in vital sign monitoring. Section III outlines the dataset used and the experimental setup designed for the study. Section IV details the proposed models for HR and BR, explaining the design and deployment on different platforms. Section V presents the results, providing insights and analysis of the findings. Lastly, Section VI concludes the work, summarizing the key contributions and suggesting avenues for future research.

II. PRINCIPLES OF THE FMCW SYSTEM FOR VITAL SIGN MONITORING

Contactless HR-BR measurement using mmWave FMCW radar detects subtle changes in the phase of the beat signal, known as the IF signal, which is linked to the micromovements of the chest wall. Radar systems rely on electromagnetic waves to detect objects, consisting of three key components: a transmitter, a receiver, and digital signal-processing modules, as shown in Fig. 2. In the FMCW radar, a frequency-modulated signal, referred to as a "chirp" is transmitted. This chirp starts at a minimum frequency, $f_{\min}$, and increases linearly to a maximum frequency, $f_{\max}$, over a time duration T_c , as shown in Fig. 3. The difference between $f_{\max}$ and $f_{\min}$ is the chirp bandwidth, represented as $B = f_{\max} - f_{\min} = \gamma T_c$, where γ is the frequency sweep rate. The transmitted chirp signal $S_{TX}(t)$ is given by

$$S_{TX}(t) = A_t \cos(2\pi f_{\min} t + \pi \gamma t^2 + \theta(t)) \quad (1)$$

where A_t is the amplitude of the transmitted signal and $\theta(t)$ is the phase noise.

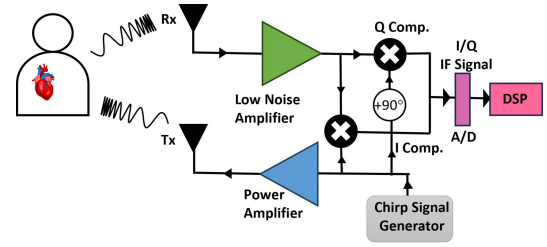


Fig. 2. mmWave FMCW radar system.

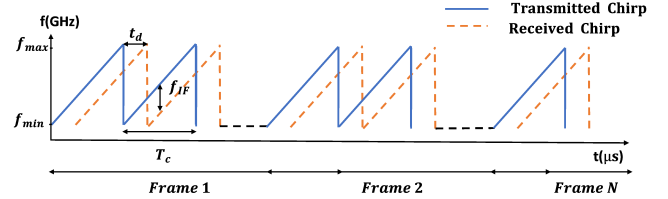


Fig. 3. Transmitted and received chirps' frequency variation in an mmWave FMCW radar.

The radar transmits these chirps sequentially with an interval T_m , which is longer than T_c , and receives the reflected signals after a time delay, $t_d = 2R/c$, where R is the distance to the target and c is the speed of electromagnetic wave in vacuum. The received signal $S_{RX}(t)$ after reflecting off the target is expressed as follows:

$$S_{RX}(t) = A_r \cos(2\pi f_{\min}(t - t_d) + \pi \gamma(t - t_d)^2 + \theta(t - t_d)) \quad (2)$$

where A_r is the amplitude of the received signal. By correlating the transmitted and received signals, the radar generates an IF signal. The IF signal's frequency corresponds to the target's distance, and its phase is sensitive to the vibrations of the chest wall caused by cardiopulmonary activity. A homodyne receiver amplifies the reflected signals, which are influenced by the object's range and radar cross-sectional area. The received reflected signal is mixed with the transmitted signal to produce the IF signal. Mathematically, the IF signal is expressed as follows:

$$S_{IF}(t) = A_t A_r \exp[j(2\pi f_{\min} t_d + 2\pi \gamma t_d t)] \quad (3)$$

To calculate the subject's range R , the equation used is

$$R = \frac{c}{2BT_c} f_{IF} \quad (4)$$

where f_{IF} is the IF signal's frequency. Additionally, the phase of the IF signal can be expressed as follows:

$$\phi(t) = \frac{4\pi f_{\min}[R + x(t)]}{c} \quad (5)$$

where $x(t)$ represents the small variations in the subject's thoracic cavity caused by cardiopulmonary activity. The IF signal's spectrogram generally follows a sawtooth waveform because the transmitted chirp has a linearly changing frequency. Mixing it with the delayed received chirp produces the IF signal that resets at the start of each new chirp, composed of multiple chirps, with one frame consisting of N chirps. The radar signal, received after the mixer circuit, consists of two

TABLE I
RADAR CONFIGURATIONS FOR THE EXPERIMENTAL SETUP

Radar Parameter	Value	Radar Parameter	Value
Start Freq., f_{min}	77 GHz	ADC Sampling freq.	6250 ksp/s
End Freq., f_{max}	81 GHz	Ramp End Time	50 μ s
ADC Start Time	6 μ s	Time Between Chirps	7 μ s
No. of ADC Samples	250	Total Frames	1200
Frame Periodicity	50 ms	Chirps per Frame, M	128
Receiver Gain	30 dB	No. of TX & RX	2 & 4

key components: in-phase (I) and quadrature (Q) components expressed as follows:

$$I(t) = A_t A_r \cos(2\pi f_{IF}t + \phi(t)) \quad (6)$$

and

$$Q(t) = A_t A_r \sin(2\pi f_{IF}t + \phi(t)). \quad (7)$$

The I/Q mixer generates these components by correlating the transmitted and received signals with two reference signals, cosine (for I) and sine (for Q). The I component captures the direct phase and amplitude of the signal, while the component Q is phase-shifted by 90° relative to the component I . The analog-to-digital converter (ADC) samples the received signal, and the discrete representation for the n th ADC sample in the m th chirp is given by

$$y[n, m] = A_t A_r \exp(j(2\pi f_{IF}nT_f + \phi(nT_f + mT_m))) \quad (8)$$

where n is the ADC sample index, m is the Chirp index, T_f is the sampling interval in fast time. This comprehensive signal model enables radar-based HR monitoring by analyzing the phase and frequency variations induced by chest wall vibrations, enabling accurate and contactless measurement of cardiopulmonary activity.

III. DATASET AND THE EXPERIMENTAL SETUP

For this study, a publicly available dataset was used, provided by the University of Twente, Netherlands [44]. This dataset, among the first of its kind, utilizes an mmWave FMCW radar to measure HR and BR under various conditions accurately. The dataset includes recordings from ten participants in various settings, such as different ranges (40, 80 cm, etc.), orientations, directions, and certain physiological states, such as those who have asthma or meditate. The dataset has been thoroughly verified against the Polar H10 sensor, provides great accuracy and dependability, making it a valuable tool for scientific investigation and analysis. The recordings were meticulously conducted in four separate 1-min sessions for each of the ten participants under varying conditions, such as different distances, orientations, and HRs (both normal and elevated). Data collection was carried out using the Texas Instruments (TI) AWR1642BOOST mmWave FMCW radar sensor module, complemented by the DCA1000 board, to capture raw ADC samples of the radar's IF signal. The specifications of the radar board are detailed in Table I.

To ensure consistent data quality, the radar system comprising the AWR1642 EVM, the DCA1000 board, and a radar holder was positioned on an office table. Adjustments were made to align the radar's antenna array with the participants' chest level. The radar was kept stationary throughout the study, with participants seated on a chair approximately 45 cm

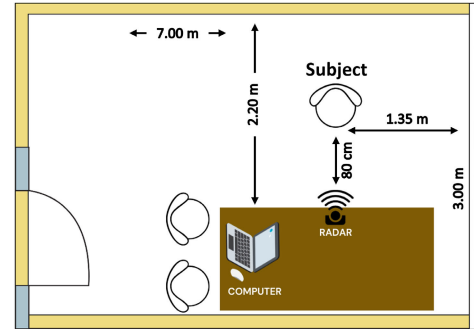


Fig. 4. Floor map: dimensions and the experimental setup.

above the ground. The radar's antennas were set at a height of about 90 cm from the floor, maintaining an 80 cm distance between the radar and the participant. Participants were instructed to minimize body movement during the sessions, but no specific instructions were provided regarding their respiration; they were neither asked to maintain a regular pattern nor adopt any abnormal breathing style. For this study, only data from all participants recorded at an 80 cm range and a 0° orientation with respect to the normal of the radar were used. The experimental setup floor map is shown in Fig. 4.

IV. PROPOSED METHOD AND ITS DEPLOYMENT

For this study, a total of 40 recordings were used, with 4 recordings from each of the 10 participants, all acquired at a range of 80 cm. To ensure robust training and evaluation, a "leave-one-experiment-out" (LOEO) strategy was employed. In this method, one experiment from all participants was set aside as the test set, while the remaining experiments were combined to form the train and validation set (8:2). This rotation was repeated until each experiment had been included in the testing phase at least once. The models were trained four times with 50 epochs each, with an early callback of a patience level of 5, to avoid overtraining. This approach ensured that the models were evaluated on entirely unseen data in every iteration, simulating real-world scenarios where the model encounters new conditions. By exposing the model to diverse training data while holding out independent test data, this strategy improved the model's ability to generalize and enhanced its reliability for broader applications. To further assess the proposed models' performance, fivefold cross-validation is also performed, offering a more stable and averaged evaluation across varying experimental conditions.

A. Preprocessing

In this study, data from the single (first) receiver were analyzed. The model was trained using a 20-s input window with a stride of 1 s. To prepare the input data, the signals from the first chirp from each frame were extracted and organized into a 2-D matrix with dimensions 400×250 , as shown in Fig. 5(a). Here, 400 represents the number of frames in the 20-s window, and 250 corresponds to the ADC samples captured in the signals. This process was repeated for the remaining 127 chirps [Fig. 5(b)], reshaping each into matrices of the same size [Fig. 5(c)]. The choice of a 20-s window was inspired by the study [36], where different window sizes (4, 8, 16, and 20 s) were tested, and the 20-s window provided the best results. Moreover, using a longer window length includes

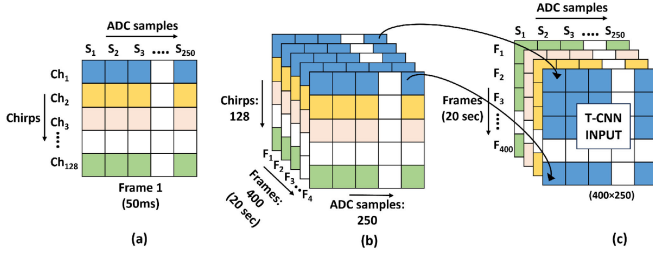


Fig. 5. Illustration of radar data organization: (a) Chirps and ADC samples in a single frame. (b) Radar data cube representing multiple frames. (c) Structure of radar data for model input preparation.

more HR and BR cycles in the data, resulting in the acquisition of higher-quality information. However, increasing the window size further would lead to a larger input feature size, which in turn increases the model parameters and the requirement for data storage, as well as decreases the number of samples in the dataset.

The IF signal comprises both in-phase (I) and quadrature (Q) components. For this study, the Q component was selected for further analysis. To ensure consistency and scale the data effectively, min–max normalization was applied to the Q component, using

$$Q_{\text{normalized}} = \frac{Q - Q_{\min}}{Q_{\max} - Q_{\min}} \quad (9)$$

where $Q_{\text{normalized}}$, Q , $Q_{\min}$, and $Q_{\max}$ are the normalized value, original value, global minimum, and global maximum of the Q component across all signals, respectively. This normalization technique used the global maximum value across all the signals and helped in normalizing the data while preserving the relative magnitudes and the small variations in the signals over time.

For the HR labels corresponding to the 20-s window, the ground truth was computed as the mean value across the window. For the BR labels, a logarithmic transformation was applied to address the wide range of values (4–40 br/min). This transformation compresses the range of values and reduces the impact of small errors (e.g., 1–2 br/min) on the relative error, thereby improving loss convergence during model training. The transformation can be mathematically given by

$$\text{BR}_{\log} = \log_e(\text{BR}) \quad (10)$$

where BR represents the original breathing rate in br/min and $\text{BR}_{\log}$ is the logarithmically transformed breathing rate. This approach ensures that the range of values is compressed, and small relative errors have a diminished effect on the overall loss function, leading to better performance during training.

B. Proposed Models

For this study, a temporal convolution (TC) was employed to address the regression task of estimating HR and BR from mmWave FMCW radar data. This section explains the concept of T-CNNs and details the proposed model architectures for HR and BR.

1) **TC Block:** A TC block works on two fundamental principles [45]. First, the network produces an output sequence that matches the length of the input sequence, ensuring temporal consistency; and second, it enforces causality, meaning that

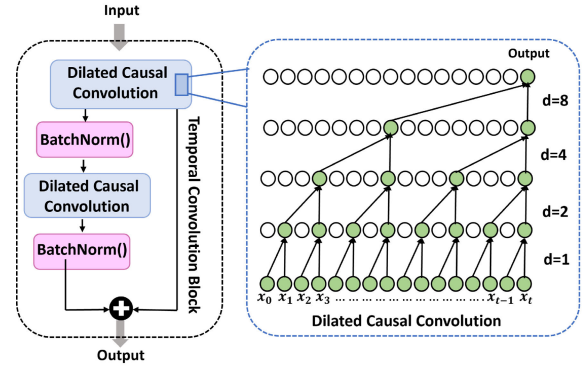


Fig. 6. TC block utilizing dilated causal convolutions. The input sequence $X = \{x_0, x_1, x_2, x_3, x_t\}$ is processed through convolutional layers with increasing dilation rates ($d = 1, 2, 4, 8$).

there is no leakage of information from future time steps to past ones, making it suitable for time-series forecasting tasks where only past and present data should inform the predictions.

Mathematically, the TC block is structured using causal convolutions, where each output at time is computed by considering only the input values up to that time, thus preventing future information from influencing the current prediction. The convolution operation can be expressed as follows:

$$y_t = \mathbf{w}^\top \mathbf{x}_t \quad (11)$$

where y_t represents the output vector at time t , $\mathbf{x}_t$ represents the input vector at time t , and $\mathbf{w}$ are the learned convolutional weight vector. To capture long-range dependencies within the time series without exponentially increasing the number of parameters, the TC block utilizes dilated convolutions. Dilation allows the receptive field to expand by skipping over values in the input sequence, which is particularly effective for capturing context over larger time spans. The dilated convolution operation is given by

$$y_t = \mathbf{w}^\top \mathbf{x}_t^{(d)} \quad (12)$$

where d is the dilation rate that plays an important role in a TC block; it introduces the gaps in the filter weights as shown in Fig. 6, thereby increasing its receptive field without increasing the model parameters.

The TC block also incorporates skip connections, which enable information to bypass intermediate layers. Skip connections make it easier for gradients to flow during training, helping the model learn more effectively and stay stable. As shown in Fig. 6, batch normalization is applied to each layer's activations, which improves the network's performance and ensures it learns reliably and consistently. In summary, the TC is a combination of causal convolutions, dilated convolutions, skip connections, and batch normalization that allows the model to effectively capture both short- and long-term temporal dependencies, making it an ideal architecture for monitoring vital signs from radar data in this study.

2) **Proposed Architecture for HR:** In this study, a T-CNN with 81 649 parameters, as shown in Fig. 7, is proposed for HR monitoring using FMCW radar data, carefully designed for computational efficiency and suitability for deployment on portable embedded systems. The model takes input data of shape (400, 250), where 400 represents the number of frames

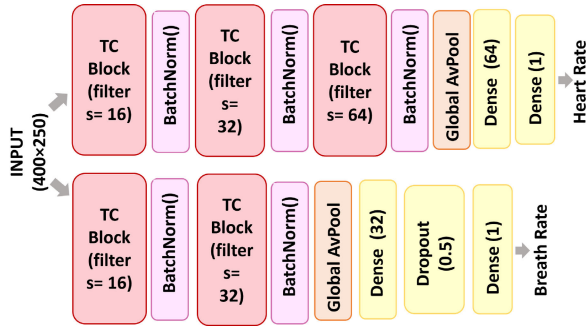


Fig. 7. Proposed model architectures for HR and BR.

and 250 corresponds to ADC samples. It consists of three TC blocks: the first block has 16 filters, a kernel size of 3, causal padding, and dilation rates of (1, 2), followed by Batch Normalization and tanh activation. The second block increases to 32 filters with dilation rates (4, 8), and the third block further scales up to 64 filters with dilation rates (16, 32). All blocks maintain causal padding, Batch Normalization, and tanh activation for consistency. After these blocks, a GlobalAveragePooling1D layer is used to reduce the dimensionality, followed by a dense layer with 64 neurons and ReLU activation, and a final dense layer with one neuron and linear activation for the regression output.

The model is optimized using mean squared error (MSE) as the loss function

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N [\hat{y}_i - y_i]^2 \quad (13)$$

where $\hat{y}_i$ is the estimated HR for the i th sample, y_i is the true HR for the i th sample, and N is the total number of samples, the Adam optimizer with a learning rate of 10^{-3} , and MAE given by

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i| \quad (14)$$

and the RMSE as the evaluation metric. Here, the absolute difference between the estimated and true values is averaged over all samples, providing an intuitive measure of accuracy.

3) Proposed Architecture for BR: For BR estimation, a similar T-CNN is proposed, with a total of 31 857 parameters, as shown in Fig. 7. The model takes input data of shape (400, 250), the same as that of the HR model. It consists of two TCN blocks: the first block has 16 filters, a kernel size of 3, causal padding, and dilation rates of (1, 2), followed by Batch Normalization and tanh activation. The second block increases the number of filters to 32, with dilation rates of (4, 8), maintaining causal padding, Batch Normalization, and tanh activation. After the TC blocks, a GlobalAveragePooling1D layer is applied to reduce dimensionality, followed by a dense layer with 32 neurons and ReLU activation, a dropout layer with a 0.5 rate to prevent overfitting, and a final dense layer with one neuron and linear activation for the BR output. The model is optimized using a modified mse loss function, defined as follows:

$$\mathcal{L} = \sum_{i=1}^N [\log(\hat{y}_i) - \log(y_i)]^2 \quad (15)$$

which can also be expressed as follows:

$$\mathcal{L} = \sum_{i=1}^N \left[\log \left(\frac{\hat{y}_i}{y_i} \right) \right]^2 \quad (16)$$

where y_i and $\hat{y}_i$ represent the true values and the predicted values, respectively. This loss function focuses on the ratio $\hat{y}_i/y_i$, ensuring the model penalizes relative errors rather than absolute differences. The logarithmic transformation makes the loss scale-invariant, treating small and large values proportionally and assigning higher penalties to errors in lower BR. The model uses the Adam optimizer for training and evaluates performance using the MAE and RMSE metrics between the original truth labels and predicted labels, balancing effective optimization with interpretable evaluation.

C. Training Phase

The model training was conducted on a high-performance system powered by an Intel Core i9-7920X processor with a base clock speed of 2.90 GHz and supported by 32 GB of RAM. For GPU acceleration, the NVIDIA TITAN Xp was utilized, offering 12 GB of dedicated memory and optimized for demanding computational tasks. The model was compiled and trained using TensorFlow GPU version 2.10.0, ensuring efficient utilization of the GPU for faster processing. Additionally, the Keras-TCN library (ver. 3.5.0) was used to build the TCNN architecture. This powerful combination of hardware and software enabled smooth, effective, and faster training.

D. Deployment on Edge Device—PYNQ-ZU

Testing was conducted on three platforms: CPU, GPU, and the embedded SoC platform PYNQ-ZU [46]. Given the complexity and time required for training the neural network, this process was carried out on the GPU to ensure efficiency and manage computational demands. However, to thoroughly evaluate the proposed models' performance across different hardware configurations, the testing was conducted on all three platforms. The PYNQ-ZU platform is based on the Xilinx Zynq UltraScale+ XCZU5EG MPSoC, featuring a quad-core ARM Cortex-A53 processor, 4GB DDR4 memory, and 256 KB on-chip memory. It includes 117K CLB LUTs, 1,248 DSP slices, and operates at a core clock speed over 450 MHz. The platform also has 32 KB L1 cache, 1 MB L2 cache, ARM Mali¹-400MP GPU, and 16GB of storage via a Micro SD card. Its TDP is 4.7 W. To evaluate the performance of the proposed model across CPU, GPU, and PYNQ-ZU, metrics such as the GFLOPS (Giga floating-point operations per second) and the inference time were analyzed.

V. RESULTS AND DISCUSSION

The performance of the proposed models for HR and BR, using the LOEO strategy, is summarized in Tables II and III, respectively, and performance for fivefold cross-validation is shown in Fig. 8. The training and validation loss curves for the LOEO and fivefold cross-validation settings for BR and HR are illustrated in Figs. 9 and 10, respectively. The MAE curves followed a similar trend and are therefore not shown for brevity. The training and validation losses initially fluctuated slightly but stabilized as training progressed. The fivefold

¹Trademarked.

TABLE II

RESULTS FOR HR ESTIMATION USING THE PROPOSED MODEL AND COMPARISON WITH THE STATE-OF-THE-ART MODELS ON THE GPU

Model	Params (M)	RMSE (bpm)	MAE (bpm)	Inference Time (ms)
ResNet1D	0.65	3.87	3.07	0.35
T-LeNet	0.51	4.46	3.74	0.30
Proposed Model	0.08	3.05	2.37	0.21

TABLE III

RESULTS FOR BR ESTIMATION USING THE PROPOSED MODEL AND COMPARISON WITH THE STATE-OF-THE-ART MODELS ON THE GPU

Model	Params (M)	RMSE (Br/min)	MAE (Br/min)	Inference Time (ms)
ResNet1D	0.65	1.62	1.07	0.19
T-LeNet	0.51	3.10	2.15	0.25
Proposed Model	0.03	1.60	1.18	0.15

RMSE and MAE Across 5-fold Cross-validation HR and BR

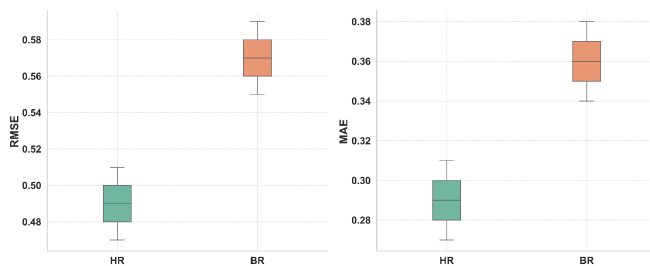


Fig. 8. Performance evaluation: RMSE and MAE across fivefold cross-validation of proposed models for HR and BR.

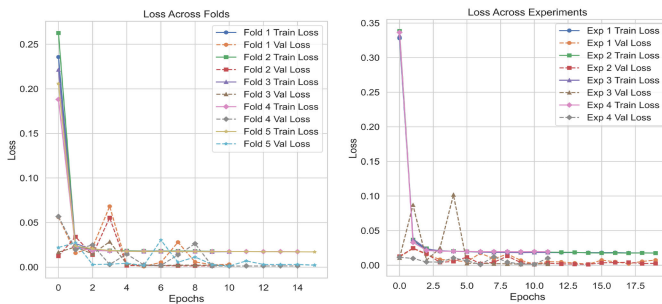


Fig. 9. Training and validation loss curves for BR estimation under fivefold cross-validation and LOEO settings.

validation led to slightly improved performance, which can be attributed to the fact that the test folds, in this case, may include data points from experiments partially seen during training. It is worth noting that the dataset used in this study has shown significant variations within the same subject across different experiments. For instance, the average HR varied by more than 6 bpm between two experiments from the same participants. Therefore, treating each experiment as an independent and unseen scenario, as done in the LOEO evaluation, provides a realistic and challenging test of generalization. This strategy ensures that the proposed models are capable of adapting to significant variations within the same individual. Furthermore, as mentioned earlier, a total of 40 recordings were considered, with ten recordings used for testing in each iteration. It was observed that more than half of

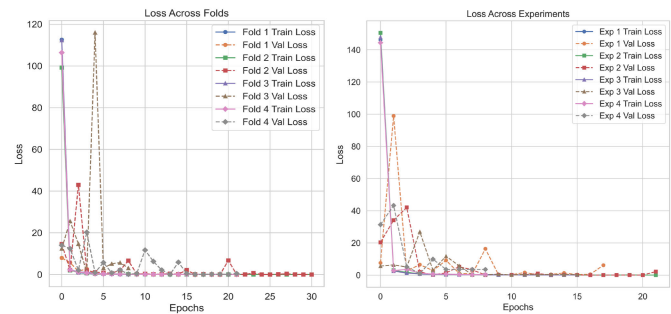


Fig. 10. Training and validation loss curves for HR estimation under fivefold cross-validation and LOEO settings.

the recordings (55%) had an error within 1 bpm, 77.5% were within 2 bpm, and 85% exhibited an error below 3 bpm.

To evaluate the effectiveness of the proposed models, their performance is also compared with the state-of-the-art models ResNet1D and T-LeNet, trained in the same LOEO fashion with the same batch size and epochs, due to the aforementioned reason of intrasubject HR variability. The performances of the ResNet1D and T-LeNet for HR and BR are shown in Tables II and III.

It should be noted that the T-LeNet is based on conventional convolutional layers without any advanced architectural enhancements, which is why it delivered the least accurate results for both HR and BR estimation. On the other hand, ResNet1D incorporates convolutional layers with residual connections, which help improve the flow of information during training and prevent issues like vanishing gradients. This gives ResNet1D an edge over the T-LeNet, resulting in better performance for both HR and BR. The proposed models take this a step further by using dilated convolutions combined with residual connections. This design not only reduces the number of trainable parameters but also improves the model's ability to capture long-range patterns in the data, which is critical for accurate HR and BR estimation. By keeping the models lightweight and efficient, they avoid overfitting while still delivering impressive results. For BR, the proposed model managed to use just 5% of the parameters required by ResNet1D while matching its performance. For HR, it used only 15% of ResNet1D's parameters and achieved even better results. The boost in HR performance can be credited to the dilated convolutions, which are particularly effective at picking up finer details in the radar signals, combined with the stability provided by residual connections. To further assess the efficacy of the proposed models, their performance is compared with recently published deep learning-based models and weighted techniques [36], [39], [40]. As illustrated in Table IV, the proposed model outperforms both the HeartBeatNet and zDAWS-HR with the zoomed-in view factor of 8 (on which it has achieved the least error [39]) in terms of MAE and parameters, as well as the inference time in the case of HeartbeatNet. The results for HeartBeatNet [36] and zDAWS-HR [39] are quoted from their respective studies and were not retrained by us; hence, some metrics are not available.

According to the study [36], the HeartBeatNet was trained and inferred on an Intel Xenon (high-end workstation). Additionally, the proposed model utilizes only 15% of the parameters compared to HeartBeatNet and 38% fewer

TABLE IV

COMPARISON OF HR ESTIMATION WITH THE EXISTING STUDIES

Study	Params (M)	MAE (bpm)	Inference Time (ms)
HeartBeatNet [36]	0.53	3.82	137.98
zDAWS-HR [39]	0.13	2.45	-
Proposed Model	0.08	2.37	0.21

TABLE V

COMPARISON OF BR ESTIMATION WITH THE EXISTING STUDIES

Study	MAE (Br/min)	Params (M)
DAWS [40]	< 3	-
zDAWS-BR [39]	1.5	0.13
Proposed Model	1.18	0.03

TABLE VI

RESULTS FOR HR AND BR ESTIMATION ACROSS DIFFERENT PLATFORMS

Metric	GPU	CPU	PYNQ-ZU
HR Inference Time (ms)	0.21	3.29	12.61
BR Inference Time (ms)	0.15	1.26	6.14
HR GFLOPS	290.32	26.57	4.833
BR GFLOPS	160.57	18.92	3.88

parameters as compared to zDAWS-HR [39], making the proposed model highly efficient.

A similar comparison is also conducted for BR estimation, which is shown in Table V. While [40] reports that 95% of respiration error estimations lie within 3 br/min, no detailed error breakdown is provided. In comparison, the proposed model not only achieves a lower overall MAE of 1.18 br/min but also exhibits a favorable cumulative error profile: 80% of recordings are within 1 br/min, 92.5% within 2 br/min, and 97.5% within 3 br/min. This distribution indicates that the majority of estimations remain highly accurate, reflecting a clear performance gain over existing methods.

The proposed models were also deployed on the PYNQ-ZU embedded board. Table VI presents a comparison of the models' evaluation performance across different platforms: PYNQ-ZU, CPU, and GPU. Although the PYNQ-ZU achieves lower GFLOPS compared to the CPU and GPU, this is because the computer system performs better than the PYNQ-ZU. On PYNQ-ZU, the proposed models have achieved an inference time of 12.6 ms per sample for HR and 6.1 ms for BR. This shows that for continuous monitoring, the inference time has been drastically reduced, making this approach an efficient and faster HR-BR monitoring system on an embedded platform.

It is important to note that the proposed models consistently delivered the same MAE and RMSE values across all platforms, regardless of hardware variations. This highlights the models' durability and scalability, making them suitable for deployment on embedded systems without compromising accuracy. The models' adaptability and dependability in practical applications are highlighted by their capacity to sustain performance across a range of hardware configurations.

VI. CONCLUSION

In conclusion, the proposed models for both HR and BR outperform both state-of-the-art models, ResNet1D and

T-LeNet, the recent HeartBeatNet model, and deep learning-based weighted schemes, showing notable improvements in MAE, RMSE, as well as in the inference time. With fewer parameters, the proposed models offer a highly efficient solution without compromising performance. Additionally, the proposed HR-BR monitoring scheme requires minimal pre-processing and eliminates the need for range bin extraction, enabling it to effectively handle small variations in subject position without compromising accuracy. Furthermore, its successful implementation on the PYNQ-ZU embedded board, along with consistent results across CPU and GPU platforms, highlights its scalability and suitability for real-world applications. This work sets the stage for future developments in radar-based vital sign monitoring, providing a balance of accuracy and efficiency in diverse portable systems.

For future work, the system can be enhanced to accommodate multiple targets, enabling simultaneous monitoring of multiple individuals. Additionally, the model can be further optimized to improve accuracy, computational efficiency, and robustness, ensuring real-time applicability and adaptability to diverse environmental conditions and variations in subject characteristics.

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REFERENCES

- [1] U. R. Acharya, K. P. Joseph, N. Kannathal, C. M. Lim, and J. S. Suri, "Heart rate variability: A review," *Med. Biol. Eng. Comput.*, vol. 44, no. 12, pp. 1031–1051, Nov. 2006, doi: [10.1007/s11517-006-0119-0](https://doi.org/10.1007/s11517-006-0119-0).
- [2] R. Avram et al., "Real-world heart rate norms in the health eHeart study," *NPJ Digit. Med.*, vol. 2, no. 1, pp. 1–10, Jun. 2019, doi: [10.1038/s41746-019-0134-9](https://doi.org/10.1038/s41746-019-0134-9).
- [3] A. M. Weissler, R. G. Peeler, and W. H. Roehll, "Relationships between left ventricular ejection time, stroke volume, and heart rate in normal individuals and patients with cardiovascular disease," *Amer. Heart J.*, vol. 62, no. 3, pp. 367–378, Sep. 1961, doi: [10.1016/0002-8703\(61\)90403-3](https://doi.org/10.1016/0002-8703(61)90403-3).
- [4] Z. Zhang, "Photoplethysmography-based heart rate monitoring in physical activities via joint sparse spectrum reconstruction," *IEEE Trans. Biomed. Eng.*, vol. 62, no. 8, pp. 1902–1910, Aug. 2015, doi: [10.1109/TBME.2015.2406332](https://doi.org/10.1109/TBME.2015.2406332).
- [5] M. Zijlmans, D. Flanagan, and J. Gotman, "Heart rate changes and ECG abnormalities during epileptic seizures: Prevalence and definition of an objective clinical sign," *Epilepsia*, vol. 43, no. 8, pp. 847–854, Aug. 2002, doi: [10.1046/j.1528-1157.2002.37801.x](https://doi.org/10.1046/j.1528-1157.2002.37801.x).
- [6] A. El Abbaoui, D. Sodoyer, and F. Elbahhar, "Contactless heart and respiration rates estimation and classification of driver physiological states using CW radar and temporal neural networks," *Sensors*, vol. 23, no. 23, p. 9457, Nov. 2023. [Online]. Available: <https://www.mdpi.com/1424-8220/23/23/9457>
- [7] I. Seflek, Y. E. Acar, and E. Yaldiz, "Small motion detection and non-contact vital signs monitoring with continuous wave Doppler radars," *Elektronika ir Elektrotechnika*, vol. 26, no. 3, pp. 54–60, Jun. 2020, doi: [10.5755/j01.eie.26.3.25810](https://doi.org/10.5755/j01.eie.26.3.25810).
- [8] K.-M. Chen, Y. Huang, J. Zhang, and A. Norman, "Microwave life-detection systems for searching human subjects under earthquake rubble or behind barrier," *IEEE Trans. Biomed. Eng.*, vol. 47, no. 1, pp. 105–114, Jan. 2000.
- [9] M. Forouzangfar, M. Mabrouk, S. Rajan, M. Bolic, H. R. Dajani, and V. Z. Groza, "Event recognition for contactless activity monitoring using phase-modulated continuous wave radar," *IEEE Trans. Biomed. Eng.*, vol. 64, no. 2, pp. 479–491, Feb. 2017.
- [10] F. JalaliBidgoli, S. Moghadami, and S. Ardalan, "A compact portable microwave life-detection device for finding survivors," *IEEE Embedded Syst. Lett.*, vol. 8, no. 1, pp. 10–13, Mar. 2016.

- [11] M. Kebe, R. Gadhafi, B. Mohammad, M. Sanduleanu, H. Saleh, and M. Al-Qutayri, "Human vital signs detection methods and potential using radars: A review," *Sensors*, vol. 20, no. 5, p. 1454, Mar. 2020, doi: [10.3390/s20051454](https://doi.org/10.3390/s20051454).
- [12] Z. Luo, Y. Liao, and M. Xing, "Target indication with FMCW-OAM radar," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023, doi: [10.1109/LGRS.2023.3314459](https://doi.org/10.1109/LGRS.2023.3314459).
- [13] M. Ambrosanio, S. Franceschini, G. Grassini, and F. Baselice, "A multi-channel ultrasound system for non-contact heart rate monitoring," *IEEE Sensors J.*, vol. 20, no. 4, pp. 2064–2074, Feb. 2020, doi: [10.1109/JSEN.2019.2949435](https://doi.org/10.1109/JSEN.2019.2949435).
- [14] S. Franceschini, M. Ambrosanio, and F. Baselice, "MUHD: A multi-channel ultrasound prototype for remote heartbeat detection," in *Proc. 13th Int. Joint Conf. Biomed. Eng. Syst. Technol.*, 2020, pp. 57–63, doi: [10.5220/0008982700002513](https://doi.org/10.5220/0008982700002513).
- [15] H.-I. Choi, W.-J. Song, H. Song, and H.-C. Shin, "Selecting target range with accurate vital sign using spatial phase coherency of FMCW radar," *Appl. Sci.*, vol. 11, no. 10, p. 4514, May 2021, doi: [10.3390/app11104514](https://doi.org/10.3390/app11104514).
- [16] H.-I. Choi, H. Song, and H.-C. Shin, "Target range selection of FMCW radar for accurate vital information extraction," *IEEE Access*, vol. 9, pp. 1261–1270, 2021, doi: [10.1109/ACCESS.2020.3043013](https://doi.org/10.1109/ACCESS.2020.3043013).
- [17] M. Alizadeh, G. Shaker, J. C. M. D. Almeida, P. P. Morita, and S. Safavi-Naeini, "Remote monitoring of human vital signs using mm-wave FMCW radar," *IEEE Access*, vol. 7, pp. 54958–54968, 2019, doi: [10.1109/ACCESS.2019.2912956](https://doi.org/10.1109/ACCESS.2019.2912956).
- [18] A. Singhal, M. Wajid, and O. Farooq, "Mitigating motion artifacts in non-contact respiratory rate measurement utilizing mm-wave FMCW radar," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2025, pp. 1–5, doi: [10.1109/ICASSP49660.2025.10888492](https://doi.org/10.1109/ICASSP49660.2025.10888492).
- [19] E. Antolinos and J. Grajal, "Comprehensive comparison of continuous-wave and linear-frequency-modulated continuous-wave radars for short-range vital sign monitoring," *IEEE Trans. Biomed. Circuits Syst.*, vol. 17, no. 2, pp. 229–245, Apr. 2023, doi: [10.1109/TBCAS.2023.3257647](https://doi.org/10.1109/TBCAS.2023.3257647).
- [20] J. Benny, P. Mahajan, S. S. Chatterjee, M. Wajid, and A. Srivastava, "Design and measurements of mmWave FMCW radar based non-contact multi-patient heart rate and breath rate monitoring system," in *Proc. IEEE Biomed. Circuits Syst. Conf. (BioCAS)*, Oct. 2023, pp. 1–5, doi: [10.1109/BioCAS58349.2023.10388660](https://doi.org/10.1109/BioCAS58349.2023.10388660).
- [21] A. Ahmad, J. C. Roh, D. Wang, and A. Dubey, "Vital signs monitoring of multiple people using a FMCW millimeter-wave sensor," in *Proc. IEEE Radar Conf. (RadarConf)*, Jun. 2018, pp. 1450–1455, doi: [10.1109/RADAR.2018.8378778](https://doi.org/10.1109/RADAR.2018.8378778).
- [22] Y. Wang, Y. Shui, X. Yang, Z. Li, and W. Wang, "Multi-target vital signs detection using frequency-modulated continuous wave radar," *EURASIP J. Adv. Signal Process.*, vol. 2021, no. 1, p. 103, Dec. 2021. [Online]. Available: <https://asp-eurasipjournals.springeropen.com/articles/10.1186/s13634-021-00812-9>
- [23] S. M. M. Islam, N. Motoyama, S. Pacheco, and V. M. Lubecke, "Non-contact vital signs monitoring for multiple subjects using a millimeter wave FMCW automotive radar," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Aug. 2020, pp. 783–786, doi: [10.1109/IMS30576.2020.9223838](https://doi.org/10.1109/IMS30576.2020.9223838).
- [24] H. Lee, B.-H. Kim, J.-K. Park, and J.-G. Yook, "A novel vital-sign sensing algorithm for multiple subjects based on 24-GHz FMCW Doppler radar," *Remote Sens.*, vol. 11, no. 10, p. 1237, May 2019. [Online]. Available: <https://www.mdpi.com/2072-4292/11/10/1237>
- [25] M. Ali, A. Elsayed, A. Mendez, Y. Savaria, and M. Sawan, "Contact and remote breathing rate monitoring techniques: A review," *IEEE Sensors J.*, vol. 21, no. 13, pp. 14569–14586, Jul. 2021, doi: [10.1109/JSEN.2021.3072607](https://doi.org/10.1109/JSEN.2021.3072607).
- [26] J. Benny, N. N. Moudhgal, M. Khan, H. Kumar Meena, M. Wajid, and A. Srivastava, "Scalable multisubject vital sign monitoring with mmWave FMCW radar and FPGA prototyping," *IEEE Sensors J.*, vol. 25, no. 2, pp. 3571–3583, Jan. 2025.
- [27] F. Chicco, S. C. Rengifo, F. X. Pengg, E. Le Roux, and C. Enz, "Power-optimized digitally controlled oscillator in 28-nm CMOS for low-power FMCW radars," *IEEE Microw. Wireless Compon. Lett.*, vol. 31, no. 8, pp. 965–968, Aug. 2021, doi: [10.1109/LMWC.2021.3092182](https://doi.org/10.1109/LMWC.2021.3092182).
- [28] X. Huang, L. Sun, T. Tian, Z. Huang, and E. Clancy, "Real-time non-contact infant respiratory monitoring using UWB radar," in *Proc. IEEE 16th Int. Conf. Commun. Technol. (ICCT)*, Oct. 2015, pp. 493–496, doi: [10.1109/ICCT.2015.7399885](https://doi.org/10.1109/ICCT.2015.7399885).
- [29] D. Schmiech, S. Müller, and A. R. Diewald, "4-channel I/Q-radar system for vital sign monitoring in a baby incubator," in *Proc. 19th Int. Radar Symp. (IRS)*, Jun. 2018, pp. 1–9, doi: [10.23919/IRS.2018.8448163](https://doi.org/10.23919/IRS.2018.8448163).
- [30] H. Lee, B.-H. Kim, J.-K. Park, S. W. Kim, and J.-G. Yook, "A resolution enhancement technique for remote monitoring of the vital signs of multiple subjects using a 24 GHz bandwidth-limited FMCW radar," *IEEE Access*, vol. 8, pp. 1240–1248, 2020, doi: [10.1109/ACCESS.2019.2961130](https://doi.org/10.1109/ACCESS.2019.2961130).
- [31] H. Abedi, A. Ansariyan, P. P. Morita, A. Wong, J. Boger, and G. Shaker, "AI-powered noncontact in-home gait monitoring and activity recognition system based on mm-wave FMCW radar and cloud computing," *IEEE Internet Things J.*, vol. 10, no. 11, pp. 9465–9481, Jun. 2023, doi: [10.1109/JIOT.2023.3235268](https://doi.org/10.1109/JIOT.2023.3235268).
- [32] Y. Pratap Singh, A. Gupta, D. Chaudhary, M. Wajid, A. Srivastava, and P. Mahajan, "Hardware deployable edge AI solution for posture classification using mmWave radar and low-computational machine learning model," *IEEE Sensors J.*, vol. 24, no. 16, pp. 26836–26844, Aug. 2024, doi: [10.1109/JSEN.2024.3416390](https://doi.org/10.1109/JSEN.2024.3416390).
- [33] B. Li, W. Li, Y. He, W. Zhang, and H. Fu, "RadarNet: Non-contact ECG signal measurement based on FMCW radar," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–9, 2024, doi: [10.1109/TIM.2024.3476545](https://doi.org/10.1109/TIM.2024.3476545).
- [34] C. Wang, T. S. Kumar, W. De Raedt, G. Camps, H. Hallez, and B. Vanrumste, "Eat-radar: Continuous fine-grained intake gesture detection using FMCW radar and 3D temporal convolutional network with attention," *IEEE J. Biomed. Health Informat.*, vol. 28, no. 2, pp. 1000–1011, Feb. 2024, doi: [10.1109/JBHI.2023.3339703](https://doi.org/10.1109/JBHI.2023.3339703).
- [35] T. Han-Trong and H. N. Viet, "An efficient heart rate measurement system using medical radar and LSTM neural network," *J. Electr. Comput. Eng.*, vol. 2022, pp. 1–11, Dec. 2022, doi: [10.1155/2022/4696163](https://doi.org/10.1155/2022/4696163).
- [36] D. Kim, J. Choi, J. Yoon, S. Cheon, and B. Kim, "HeartBeatNet: Enhancing fast and accurate heart rate estimation with FMCW radar and lightweight deep learning," *IEEE Sensors Lett.*, vol. 8, no. 4, pp. 1–4, Apr. 2024, doi: [10.1109/LSSENS.2024.3378199](https://doi.org/10.1109/LSSENS.2024.3378199).
- [37] Y. Chen, J. Yuan, and J. Tang, "A high precision vital signs detection method based on millimeter wave radar," *Sci. Rep.*, vol. 14, no. 1, p. 25535, Oct. 2024, doi: [10.1038/s41598-024-77683-1](https://doi.org/10.1038/s41598-024-77683-1).
- [38] W. Liu, S. Zhang, J. Yang, and M. Wang, "Human vital signs detection based on millimeter wave radar and digital filtering," in *Proc. IEEE Int. Conf. Image Process. Comput. Appl. (ICIPCA)*, Aug. 2023, pp. 569–574, doi: [10.1109/ICIPCA59209.2023.10257847](https://doi.org/10.1109/ICIPCA59209.2023.10257847).
- [39] H.-Y. Chang, Y.-Y. Chen, and W.-H. Chung, "Two-stage zoom FFT-enhanced deep learning-aided weighted scheme for wireless vital sign estimation using mmWave FMCW radar," *IEEE Sensors Lett.*, vol. 8, no. 7, pp. 1–4, Jul. 2024. [Online]. Available: <https://ieeexplore.ieee.org/document/10564177/>
- [40] H.-Y. Chang, C.-H. Hsu, and W.-H. Chung, "Fast acquisition and accurate vital sign estimation with deep learning-aided weighted scheme using FMCW radar," in *Proc. IEEE 95th Veh. Technol. Conf. (VTC-Spring)*, Jul. 2022, pp. 1–6, doi: [10.1109/VTC2022-Spring54318.2022.9860799](https://doi.org/10.1109/VTC2022-Spring54318.2022.9860799).
- [41] Z. Wang, W. Yan, and T. Oates, "Time series classification from scratch with deep neural networks: A strong baseline," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, May 2017, pp. 1578–1585, doi: [10.1109/IJCNN.2017.7966039](https://doi.org/10.1109/IJCNN.2017.7966039).
- [42] R. Ganiga, M. S. N., W. Choi, and S. Pan, "ResNet1D-based personal identification with multi-session surface electromyography for electronic health record integration," *Sensors*, vol. 24, no. 10, p. 3140, May 2024. [Online]. Available: <https://www.mdpi.com/1424-8220/24/10/3140>
- [43] H. I. Fawaz, G. Forestier, J. Weber, L. Idoumghar, and P.-A. Müller, "Deep learning for time series classification: A review," *Data Mining Knowl. Discovery*, vol. 33, no. 4, pp. 917–963, Mar. 2019. [Online]. Available: <https://link.springer.com/10.1007/s10618-019-00619-1>
- [44] E. Sadeghi, K. Skurule, A. Chiumento, and P. Havinga, "Comprehensive mm-wave FMCW radar dataset for vital sign monitoring: Embracing extreme physiological scenarios," 2024, *arXiv:2405.12659*.
- [45] S. Bai, J. Zico Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," 2018, *arXiv:1803.01271*.
- [46] *Aup Pynq-Zu*. Accessed: Nov. 18, 2024. [Online]. Available: <https://www.amd.com/en/corporate/university-program/aup-boards/pynq-zu.html>